

NTHU OS Lab 4 Report

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Fairness

Using the default Round-Robin (RR) scheduling, the output patterns depend on the order of thread creation. When one producer is scheduled first, it tends to fill the buffer completely, causing the empty semaphore to reach zero. This blocks subsequent producers from executing until consumers increment the empty semaphore by consuming items. As a result, the outputs often follow sequences like “0123456789...” or “ABCDEFGHI...”, reflecting the initial scheduling order.

Fairness in output can be defined as the data from both producers alternating evenly. However, with default RR scheduling, fairness is not guaranteed, as the initial thread creation order heavily influences the pattern. This can lead to one producer dominating and the other experiencing starvation. To address this, the producers can explicitly call `ThreadYield` after producing each item, allowing the scheduler to alternate between producers. This ensures that both contribute to the output evenly, resolving the starvation problem and creating a fairer output pattern.

SDCC Compilation

Compilation. The following screenshot (Fig. 1) shows the compilation of the codes provided, using the `Makefile` script.

```
(base) kaiyuanjeng@Kristans-MacBook-Air my_ppc4 % ls
Makefile      preemptive.c  preemptive.h  test3threads.c
(base) kaiyuanjeng@Kristans-MacBook-Air my_ppc4 % make clean
rm *.hex *.ihx *.lnk *.lst *.map *.mem *.rel *.rst *.sym *.asm *.lk
rm: *.hex: No such file or directory
rm: *.ihx: No such file or directory
rm: *.lnk: No such file or directory
rm: *.lst: No such file or directory
rm: *.map: No such file or directory
rm: *.mem: No such file or directory
rm: *.rel: No such file or directory
rm: *.rst: No such file or directory
rm: *.sym: No such file or directory
rm: *.asm: No such file or directory
rm: *.lk: No such file or directory
make: *** [clean] Error 1
(base) kaiyuanjeng@Kristans-MacBook-Air my_ppc4 % make
sdcc -c test3threads.c
sdcc -c preemptive.c
preemptive.c:110: warning 85: in function ThreadCreate unreferenced function argument : 'fp'
sdcc -o test3threads.hex test3threads.rel preemptive.rel
(base) kaiyuanjeng@Kristans-MacBook-Air my_ppc4 % ls
Makefile      preemptive.rel  test3threads.hex  test3threads.rel
preemptive.asm preemptive.rst  test3threads.lk   test3threads.rst
preemptive.c  preemptive.sym  test3threads.lst  test3threads.sym
preemptive.h  test3threads.asm test3threads.map
preemptive.lst test3threads.c  test3threads.mem
(base) kaiyuanjeng@Kristans-MacBook-Air my_ppc4 % |
```

Figure 1: Compilation

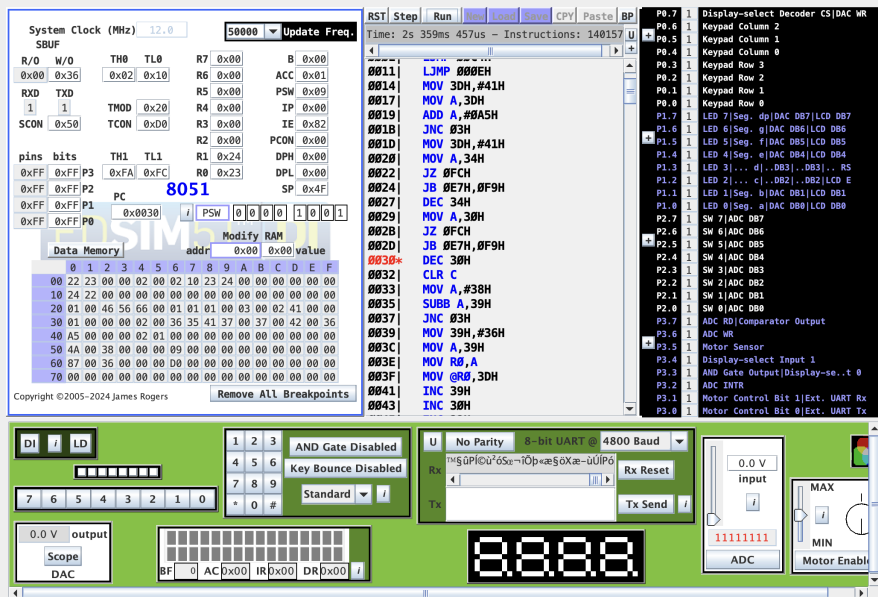


Figure 2: Semaphore Decrease

EdSim51 Simulation

Producer. The following screenshots (Fig. 2 and Fig. 3) show the changing of Semaphore.

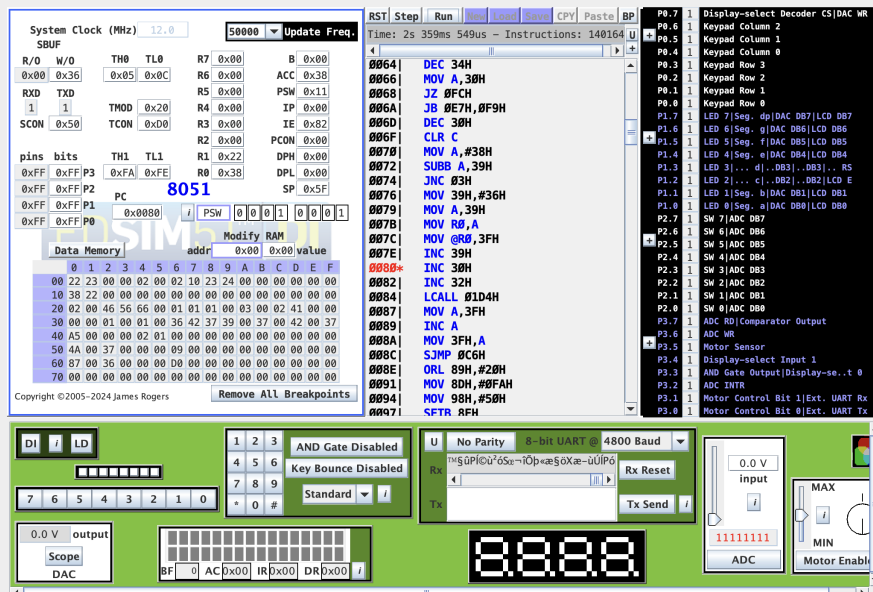


Figure 3: Semaphore Increase